PROJECT REPORT

TASK 30: EXPLORATORY DATA ANALYSIS USING PYTHON

1. Abstract

This project focuses on performing Exploratory Data Analysis (EDA) using Python, Pandas, NumPy, and Matplotlib.

A student performance dataset containing academic information is analyzed to understand its structure, identify data-quality issues, calculate statistical measurements, and discover meaningful patterns.

The project combines data exploration, cleaning, descriptive statistics, correlation analysis, and visualization into one complete workflow.

Five visualizations are generated to examine the distribution of marks, department-wise performance, relationships between numerical variables, and correlation patterns.

2. Introduction

Exploratory Data Analysis is an important step in the data analysis and machine-learning process.

It involves examining a dataset to understand its structure, characteristics, distributions, relationships, and potential data-quality issues.

EDA helps analysts identify missing information, duplicate records, unusual values, and meaningful patterns before applying statistical or machine-learning techniques.

Python libraries such as Pandas, NumPy, and Matplotlib provide useful functions for performing these operations efficiently.

This project demonstrates a basic EDA workflow using a student performance dataset.

3. Objectives

The main objectives of this project are:

- To understand the structure of a dataset.
- To inspect columns, records, and data types.
- To identify and handle missing values.
- To identify and remove duplicate records.

- To calculate descriptive statistical measurements.
- To create at least four meaningful visualizations.
- To analyze relationships between numerical variables.
- To identify at least five meaningful observations.
- To prepare and export a cleaned dataset.

4. Tools and Technologies

Python: Used as the primary programming language.

Pandas: Used for dataset loading, exploration, cleaning, grouping, and statistical calculations.

NumPy: Used for numerical operations and missing-value representation.

Matplotlib: Used to generate histograms, bar charts, scatter plots, box plots, and heatmaps.

Google Colab: Used as the Jupyter Notebook development environment.

CSV: Used to store the original dataset, cleaned dataset, and analysis summaries.

5. Dataset Description

The project uses a manually created student performance dataset containing 30 unique student records.

The original dataset includes one additional duplicate record, making 31 rows.

The dataset contains six columns:

Column	Description
student_id	Unique student identifier
gender	Student gender category
department	Academic department
study_hours	Hours spent studying
attendance_pct	Attendance percentage
marks	Examination marks

The dataset includes three academic departments:

1. Computer Science (CSE)
2. Electronics and Communication (ECE)
3. Information Technology (IT)

The dataset is fictional and created for educational and analytical purposes.

6. Methodology

The project follows these stages:

1) Import the required Python libraries.
2) Create the student performance dataset.
3) Export and load the original CSV file.
4) Inspect dataset dimensions and column names.
5) Examine data types and dataset information.
6) Identify missing values.
7) Identify and remove duplicate records.
8) Handle missing numerical and categorical values.
9) Verify the cleaned dataset.
10) Calculate descriptive statistical measurements.
11) Generate department-wise performance summaries.
12) Create five meaningful visualizations.
13) Calculate correlations between numerical variables.
14) Identify observations from the analysis.
15) Export the cleaned dataset and summary files.

7. Dataset Exploration

The original dataset is examined using Pandas functions such as:

- `shape`
- `columns`
- `dtypes`
- `head()`
- `tail()`
- `info()`
- `describe()`

These functions provide information about the dataset's dimensions, column names, data types, records, and statistical characteristics.

The original dataset contains 31 rows and six columns.

8. Data Cleaning

Data cleaning is performed to improve dataset quality before analysis.

8.1 Duplicate Removal

The original dataset contains one duplicate student record.

Pandas `duplicated()` is used to identify repeated student IDs, and `drop_duplicates()` is used to remove the duplicate.

Results:

- Original records: 31
- Duplicate records removed: 1
- Records after duplicate removal: 30

8.2 Missing-Value Analysis

Four missing cells are identified:

Column	Missing Values
department	1
study_hours	1
attendance_pct	1
marks	1
Total	4

8.3 Missing-Value Treatment

Median imputation is used for numerical columns.

Column	Median Used
study_hours	4.0
attendance_pct	88.0
marks	73.0

The missing department value is represented by `Unknown` rather than assuming an unsupported department.

After cleaning, the dataset contains zero missing cells and zero duplicate student IDs.

9. Descriptive Statistical Analysis

Descriptive statistics are calculated for study hours, attendance percentage, and examination marks.

Measurement	Study Hours	Attendance	Marks
Mean	4.35	85.77	72.83
Median	4.00	88.00	73.00
Minimum	1.00	65.00	48.00
Maximum	8.50	99.00	97.00

The mean represents the average value, while the median represents the middle observation.

Standard deviation measures how widely the values are spread around their mean.

These measurements help understand the overall characteristics of the dataset.

10. Department-wise Performance Analysis

Department-wise statistics are calculated using Pandas grouping and aggregation functions.

The student whose department is recorded as `Unknown` is retained in the cleaned dataset but excluded from the department comparison.

Department	Students	Average Marks
CSE	9	72.56
ECE	10	69.90
IT	10	78.10

IT has the highest average examination marks among the three identified departments.

ECE has the lowest average marks in this sample dataset.

11. Data Visualizations

Five visualizations are generated using Matplotlib.

11.1 Histogram of Student Marks

A histogram is created using `plt.hist()`.

It represents the distribution of examination marks and helps identify the concentration of student scores.

Insert the actual marks_distribution.png image here.

11.2 Bar Chart of Average Marks by Department

A bar chart is created using `plt.bar()`.

It compares average examination marks across CSE, ECE, and IT departments.

Insert the actual average_marks_by_department.png image here.

11.3 Scatter Plot of Study Hours and Examination Marks

A scatter plot is generated using `plt.scatter()`.

Study hours are represented on the X-axis, while examination marks are represented on the Y-axis.

A linear trend line is added to visualize the direction of the relationship.

Insert the actual study_hours_vs_marks.png image here.

11.4 Box Plot of Marks by Department

A box plot is created using `plt.boxplot()`.

It compares the median, spread, and variation of marks across departments.

Insert the actual marks_by_department_boxplot.png image here.

11.5 Correlation Heatmap

A correlation heatmap is generated using Matplotlib.

It displays the relationships among study hours, attendance percentage, and examination marks.

Insert the actual correlation_heatmap.png image here.

12. Correlation Analysis

Pearson correlation is calculated to understand the linear relationships between numerical variables.

The results are:

Variable Relationship	Correlation
Study hours and marks	0.9256
Attendance and marks	0.9320
Study hours and attendance	0.9060

All three relationships show strong positive linear associations in the cleaned sample dataset.

This indicates that higher study hours and attendance are associated with higher examination marks in this illustrative dataset.

Correlation does not establish causation.

13. Key Observations

The EDA process provides the following observations:

1. The original dataset contains 31 rows, including one duplicate record. After cleaning, 30 unique student records remain.
2. Four missing cells were identified and handled using appropriate replacement methods.
3. IT has the highest department-wise average marks of 78.10.
4. ECE has the lowest department-wise average marks of 69.90.
5. Study hours and examination marks have a strong positive correlation of approximately 0.9256.
6. Attendance and examination marks have a strong positive correlation of approximately 0.9320.
7. The average examination mark is 72.83, while the median is 73.00.
8. Average attendance is approximately 85.77%

These findings are based on the actual sample dataset.

14. Importance of Exploratory Data Analysis

EDA helps understand data before applying advanced statistical methods or machine-learning algorithms.

It enables analysts to:

- Understand dataset structure.
- Identify data-quality problems.
- Explore numerical and categorical variables.
- Understand distributions and relationships.
- Discover meaningful patterns.
- Support analytical decisions using evidence.

EDA is an important step in preparing reliable datasets for further analysis.

15. Applications

Exploratory Data Analysis is useful in:

- Student performance analysis.
- Employee performance analysis.
- Sales and business analytics.
- Customer behavior analysis.
- Healthcare data analysis.
- Financial data analysis.
- Machine-learning data preparation.
- Academic research.

16. Limitations

The project uses a small, manually created dataset of 30 fictional student records.

The data demonstrates basic EDA concepts but does not represent actual student performance.

The missing-value treatment uses simple median imputation, and correlation analysis only identifies linear associations.

Therefore, the results should not be treated as evidence of real-world causal relationships.

17. Future Enhancements

The project could be extended by:

- Using larger real-world student performance datasets.
- Performing more detailed statistical tests.
- Applying advanced outlier detection.
- Creating interactive visualizations.
- Comparing additional academic characteristics.
- Applying regression and predictive modelling.
- Developing an interactive EDA dashboard.

18. Conclusion

This project demonstrates a complete basic Exploratory Data Analysis workflow using Python, Pandas, NumPy, and Matplotlib.

It combines dataset exploration, missing-value handling, duplicate removal, descriptive statistics, visualizations, and correlation analysis.

Five visualizations are created to identify patterns and relationships in student performance data.

The analysis identifies meaningful department-wise differences and strong positive associations between study hours, attendance, and examination marks.

The project provides practical experience in preparing, analyzing, and interpreting structured data before machine-learning applications.

19. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. NumPy Documentation: https://numpy.org/doc/
3. Matplotlib Documentation: https://matplotlib.org/stable/

4. Pandas DataFrame Documentation: https://pandas.pydata.org/docs/reference/frame.html
5. Matplotlib Visualization Documentation: https://matplotlib.org/stable/plot_types/index.html

20. Project Links

GitHub Repository: https://github.com/venky1318/Exploratory_Data_Analysis